EXECUTIVE HEALTHCARE PERFORMANCE ANALYSIS

Green Valley Medical Center

Clinical Operations, Departmental Performance & Financial Risk Review

PREPARED BY

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TOOLS

Microsoft Excel • Power Query

Power Pivot • DAX

DATE • VERSION

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Portfolio Disclaimer: This project was created for educational and portfolio purposes using a fictional healthcare dataset. Green Valley Medical Center, all clinicians, patient records, and associated information are fictional and do not represent any real healthcare organisation or individual.

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EXECUTIVE SUMMARY

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Problem

Green Valley Medical Center had no structured way to compare department performance, evaluate treatment effectiveness, or track whether rising costs were matched by better patient outcomes, leaving staffing, treatment, and budget decisions without a shared evidence base.

Dataset

400 patient admissions across seven departments, covering department, treatment type, physician, age, gender, treatment cost, length of stay, and recovery score, with Age Group and Cost Category added as calculated fields. Institutional baseline: 75.3 recovery score, $5,052 average cost, 5.4-day average stay.

Top Findings

- Cardiology combines the hospital's lowest recovery score (72.9) with its highest treatment cost ($5,255) — its largest combined clinical and financial risk.
- The Observation pathway scores 2.5 points below baseline across 102 admissions, the hospital's weakest treatment track.
- An 8.7-point recovery gap separates the highest- and lowest-scoring physicians (Dr Johnson 78.5 vs. Dr Smith 69.8).
- Adults aged 30–44 score 2.1 points below baseline — the lowest-performing major age cohort — despite the lowest average cost of any age band.

Recommendations

Review Cardiology's operational model within 60 days; set a 48-hour escalation policy for Observation patients; place Dr Smith and Dr Rodriguez in structured peer review; investigate the adult cohort's care pathway; track every high-cost admission against its recovery outcome; and document Dermatology and Oncology's practices for replication elsewhere. Full detail and targets in Section 5C.

Sample-size note: Infant (n=2) and Other-gender (n=5) figures appear in this report for completeness but are too small to draw conclusions from; they are excluded from the findings above.

1. EXECUTIVE OVERVIEW & BASELINES

Background

Green Valley Medical Center collects patient data across seven departments but previously had no structured way to analyse those records for leadership decisions. This report uses a new Microsoft Excel analytics system to review 400 recent admissions. The goal is to give hospital executives a clear picture of where care is working well and where it is not, using facts rather than operational assumptions.

Problem this analysis solves

Without a shared analytical view, hospital administrators have found it difficult to compare department performance, track which treatments are producing results, or spot where costs are rising without a matching improvement in patient outcomes. Day-to-day decisions about staffing, treatment priorities, and budgets are being made without the full picture.

What this report covers

- How each of the seven departments scores on patient recovery, relative to the hospital average
- Which treatment types produce the best and worst results per dollar spent
- How individual physicians compare on patient outcomes, with patient volume for context
- Which age groups are falling below the hospital average and what may be driving that
- Where the hospital is spending the most and whether that spend generates proportionate results
- Six specific recommendations with targets and timelines leadership can act on immediately

Data preparation

The dataset covers 400 patient admissions. Each record includes the patient's department, treatment type, assigned physician, age, gender, treatment cost, length of stay, and recovery score. Two calculated fields were added: Age Group (seven life-stage cohorts) and Cost Category (Low / Medium / High). All 400 records were checked for duplicates and inconsistencies before analysis.

Methodology note: all comparative figures in this report, including cost-per-recovery-point, are aggregated at the group level (department, treatment type, physician, or age cohort) — calculated as that group's average cost divided by its average recovery score, not as a per-patient ratio averaged afterward.

Data quality note

No duplicates were found. All categorical fields were standardised before analysis. Recovery scores range from 0 to 100. Cost category bands: Low below $2,500 / Medium $2,500 to $8,000 / High above $8,000. Age Group bands: Infant under 1 / Child 1–12 / Teenager 13–17 / Youth 18–29 / Adult 30–44 / Middle-Aged 45–64 / Elderly 65 and over.

Institutional KPI Baseline

These five figures establish the hospital's baseline across all 400 patients. Every departmental, physician, and treatment comparison in this report is measured against them.

KPI	Value	What it means for leadership
Total patient cohort	400 admissions	Full operational volume across 7 departments
Recovery benchmark	75.3 / 100	Facility-wide baseline for all comparisons
Average treatment cost	$5,052	Per-admission expenditure baseline
Average hospital stay	5.4 days	Inpatient efficiency baseline
Average patient age	45 years	Skews older; Elderly is the largest cohort

What these numbers mean

A recovery score of 75.3 is the hospital-wide average, not a target. Departments and physicians above this are performing above average within this dataset; those below may warrant review. The $5,052 average cost covers a wide range: treatment-type averages run from $1,080 to $12,619, and individual high-cost admissions run higher still. The analysis below shows whether those cost differences are associated with proportionate recovery improvements.

Methodology & limitations

All figures in this report are descriptive averages drawn directly from the 400-admission dataset; no statistical significance testing was applied. The recovery score is used as supplied in the source records (a 0–100 index) and was not independently validated. Small differences — roughly a point or less — sit within normal variation and should be read as broadly level rather than as meaningful gaps.

Most importantly, recovery scores are not adjusted for case mix. Departments, physicians, and age groups treat patients of differing severity, so a lower score may reflect a sicker patient population rather than weaker care. For this reason, every department-, physician-, and cohort-level finding below should be read as an area flagged for review, not as a proven performance gap. The workflow and peer reviews recommended later are the mechanism for separating a genuine process issue from a case-mix effect.

2. CLINICAL OUTCOMES & DEPARTMENTAL PERFORMANCE

Department Performance Chart Snapshot

Figure 1 — Average Recovery Score vs. Hospital Stay Days by Department

Figure 1: *Dermatology (76.6) and General Medicine (76.6) lead all departments; Cardiology (72.9) sits 2.4 points below the hospital baseline of 75.3.*

Recovery scores vs. hospital baseline (75.3)

The chart above ranks each department by average recovery score. Cardiology is the only department with a gap large enough to warrant closer operational review, combining the hospital's lowest recovery score with its highest treatment cost.

Department	n	Recovery	vs. Avg	Avg Cost	Rating
Dermatology	64	76.6	+1.3	$4,728	High performing
General Medicine	54	76.6	+1.2	$5,157	High performing
Orthopedics	54	75.5	+0.2	$4,931	Stable
Pediatrics	59	75.5	+0.1	$5,588	Stable
Oncology	59	75.1	-0.3	$4,646	Slightly below average
Neurology	57	75.0	-0.4	$5,104	Slightly below average
Cardiology	**53**	**72.9**	**-2.4**	**$5,255**	**Needs immediate review**

Cardiology: the hospital's highest financial and clinical risk

Cardiology admissions cost an average of $5,255 but produce a recovery score of 72.9 — 2.4 points below the hospital average. Dermatology achieves a higher recovery score (76.6) while costing $527 less per patient. This pattern suggests that Cardiology may be consuming more resources without a proportionate outcome advantage. Oncology shows the opposite pattern: near-average-to-above-average recovery (75.1) at one of the hospital's lowest departmental costs ($4,646).

What leadership should do

Cardiology needs a formal workflow review within 60 days. Focus on how quickly patients are stabilised on admission, how discharge decisions are made, and whether staffing levels match patient complexity. Dermatology's results — high recovery at low cost and the hospital's shortest average stay of 4.7 days — should serve as the internal reference point.

Operational efficiency: stay duration vs. recovery

Short stays are not always better. Dermatology achieves the hospital's highest recovery score with the shortest average stay. Cardiology combines a prolonged stay with the lowest recovery score — the only department failing on both measures simultaneously.

Department	Avg Stay	Recovery	Efficiency read
Dermatology	4.7 days	76.6	Best overall — short stay, highest recovery
Oncology	4.8 days	75.1	Short stay, solid recovery
Pediatrics	5.4 days	75.5	Near-average stay, above-average recovery
Cardiology	**5.6 days**	**72.9**	**Longest active stay, lowest recovery**
General Medicine	5.9 days	76.6	Longest stay but strong recovery — capacity risk

Note: patients staying 12+ days averaged 77.3 recovery versus 73.8 for 1–2 day stays. The hospital should review whether some patients are being discharged too early.

3. TREATMENTS, PHYSICIANS & DEMOGRAPHICS

Treatment pathway performance

Treatment Pathway Chart Snapshot

Figure 2 — Treatment Type Efficacy vs. Facility Baseline

Figure 2: *Surgery leads on recovery (76.8) at the highest cost ($12,619). Observation scores lowest (72.9) at the lowest cost ($1,080). Therapy offers the best balance: 76.1 recovery for $3,954.*

Treatment	n	Avg Cost	Recovery	vs. Avg	Cost per recovery pt*
Surgery	97	$12,619	76.8	+1.5	$164.22
Therapy	99	$3,954	76.1	+0.8	$51.96
Medication	102	$2,891	75.7	+0.3	$38.21
Observation	**102**	**$1,080**	**72.9**	**-2.5**	**$14.82**

** Cost per recovery point = average cost divided by average recovery score. Lower values indicate better cost efficiency.*

Observation: the hospital's weakest treatment track

Patients on the Observation pathway average 72.9 — 2.5 points below the hospital baseline — across 102 admissions, tying it for the highest treatment volume. The lower scores may indicate that some patients remain in a holding pattern longer than is optimal and could benefit from earlier movement to an active treatment track. Surgery costs 11.7 times more than Observation but improves recovery by only 4.0 points.

What leadership should do

The hospital should require that stable Observation patients are reviewed for escalation to an active treatment within 24 to 48 hours of admission. Therapy handles a similar volume (99 cases) and produces a 76.1 recovery score at $3,954.

Physician performance

Physician Performance Chart Snapshot

Figure 3 — Average Patient Recovery Score by Physician

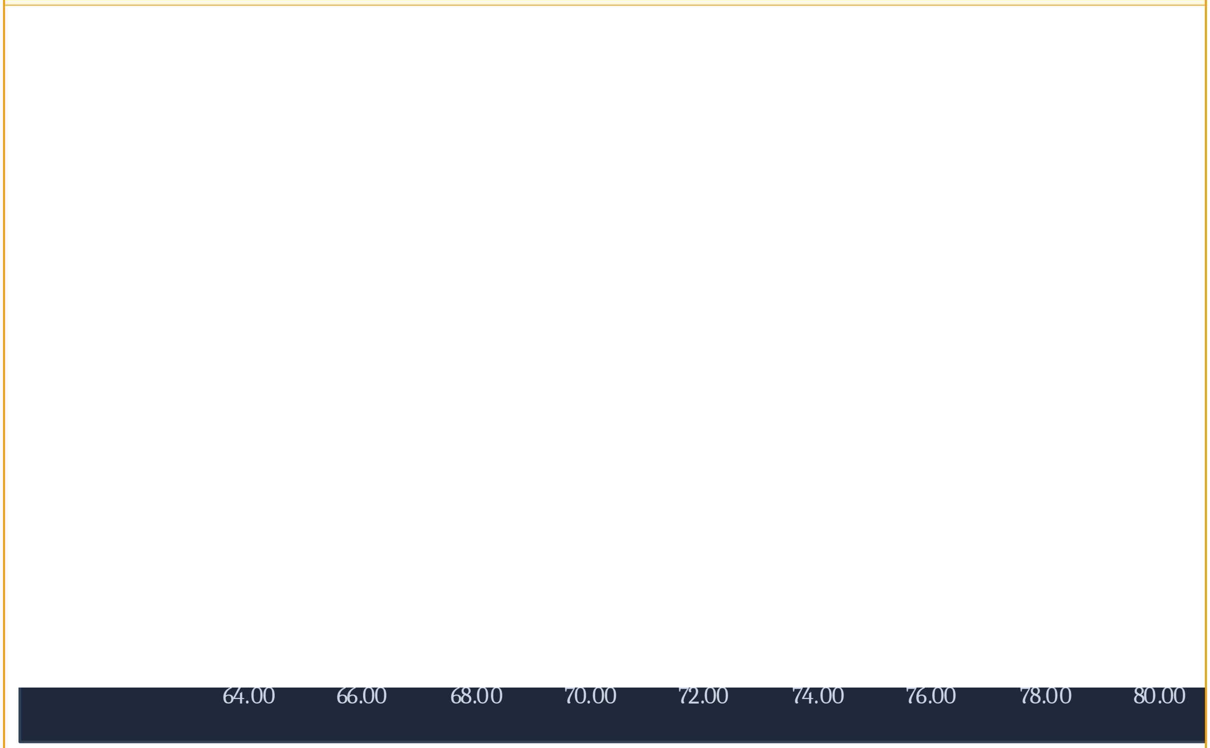

Figure 3: *Dr Johnson leads at 78.5 across 47 patients. Dr Smith records 69.8 across 27 patients — the largest gap below the hospital baseline of any physician.*

Physician	n	Recovery	vs. Avg	Avg Cost	Status
Dr Johnson	47	78.5	+3.2	$5,812	Top quartile
Dr Miller	39	77.2	+1.9	$6,379	High
Dr Jones	43	76.9	+1.6	$5,768	High
Dr Garcia	41	76.8	+1.5	$5,458	Stable
Dr Williams	32	76.4	+1.0	$4,990	Stable
Dr Brown	31	75.3	-0.1	$4,616	At average
Dr Davis	47	74.9	-0.5	$3,619	Slightly below
Dr Martinez	38	73.5	-1.8	$3,696	Below average
Dr Rodriguez	**55**	**72.8**	**-2.6**	**$4,694**	**Needs review**
Dr Smith	**27**	**69.8**	**-5.6**	**$5,756**	**Needs review**

Dr Smith and Dr Rodriguez: two physicians for structured support

Dr Smith's patients average 69.8 across 27 admissions — 5.6 points below the hospital average. His average treatment cost of $5,756 is the second-highest among all physicians. Dr Rodriguez sees more patients than anyone else (55 cases) at 72.8 — his below-average results affect more patients than anyone else's.

What leadership should do

Pair Dr Smith and Dr Rodriguez with Dr Johnson (78.5, n=47) and Dr Jones (76.9, n=43) in a structured peer review. Success metric: all physicians reach 73.0 or above within three reporting cycles.

Patient demographics

Gender Distribution Chart Snapshot

Figure 4 — Patient Volume by Gender Demographic

Figure 4: *Gender distribution is balanced (Male 206, Female 189). Recovery scores for Male (75.6) and Female (75.3) are comparable.*

Gender	Volume	Recovery	vs. Avg
Male	206 (51.5%)	75.6	+0.2
Female	189 (47.3%)	75.3	-0.1
Other*	5 (1.3%)	68.6	-6.7

** Other n=5 — too small a sample to draw conclusions from.*

Age Group Performance Chart Snapshot

Figure 5 — Patient Recovery Index by Age Group

Figure 5: *Adults (30–44) score lowest of any major cohort at 73.2. Middle-Aged (76.3) and Youth (76.2) lead all major groups.*

Age group	n	Recovery	vs. Avg	Avg cost
Infant †	2	90.5	+15.2	$9,487
Child	57	75.0	-0.4	$5,405
Teenager	30	73.9	-1.4	$6,126
Youth	65	76.2	+0.9	$5,263
Adult	**71**	**73.2**	**-2.1**	**$3,919**
Middle-Aged	74	76.3	+1.0	$5,014
Elderly	101	75.9	+0.6	$5,133

† Infant n=2 — too small a sample to report as a reliable benchmark.

Adults aged 30–44: the hospital's lowest major age group

Adults between 30 and 44 average 73.2 across 71 admissions — 2.1 points below the hospital average and 3.1 points below Middle-Aged patients (76.3) — despite the lowest average treatment cost of any age band at $3,919. This pattern may indicate that adults are more often placed on lower-intensity pathways, or that other unmeasured factors are affecting outcomes for this cohort.

4. FINANCIAL RISK & COST ANALYSIS

Spending more does not reliably produce better patient outcomes at Green Valley. This chapter examines where the money goes and what return it generates in recovery score terms.

Cost Category Chart Snapshot

Figure 6 — Patient Distribution by Financial Cost Category

Figure 6: *Low cost accounts for 43% of admissions. High cost (20%) produces stronger recovery (77.7) but at a 10.6x cost ratio.*

Cost category breakdown

Category	n	Share	Recovery score	Avg cost
High cost	81	20.3%	77.7	$13,877
Medium cost	149	37.3%	76.0	$4,528
Low cost	170	42.5%	73.6	$1,305

Cost vs. recovery quadrant

The four quadrants below show where each department sits when cost and recovery are plotted together. Only Cardiology sits in the high-cost, low-recovery zone.

High value zone *Low cost, high recovery* Dermatology: $4,728 / 76.6 Oncology: $4,646 / 75.1	**High cost, high recovery** *Spending more, getting results* General Medicine: $5,157 / 76.6 Pediatrics: $5,588 / 75.5
Average cost, average recovery *Stable — no immediate concern* Orthopedics: $4,931 / 75.5 Neurology: $5,104 / 75.0	**High cost, low recovery** *Needs immediate review* Cardiology: $5,255 / 72.9

The cost gap does not match the outcome gap

High-cost admissions average $13,877 versus $1,305 for low-cost — a 10.6x difference — yet the recovery gap is only 4.1 points. Therapy achieves nearly the same recovery as Surgery (76.1 vs. 76.8) at less than one-third of the cost.

What leadership should do

Track every high-cost admission monthly against its actual recovery score. Any case scoring below 75.3 should be reviewed at departmental level, with Cardiology reviewed first. Ask Dermatology and Oncology to document what lets them achieve strong recovery at low cost, then share those practices with Cardiology and General Medicine.

5A. STRATEGIC INSIGHTS SUMMARY

Seven findings from this analysis deserve direct attention from leadership. The first four require action now.

#	Finding	Key metric	Priority
1	**Cardiology is the hospital's largest combined cost and clinical risk**	**Score 72.9 | Cost $5,255 | Stay 5.6d**	**Immediate**
2	**Observation patients score 2.5 points below average across 102 cases**	**Score 72.9 | n=102 | -2.5 pts**	**High**
3	**Surgery costs 11.7x more than Observation for 4.0 additional recovery points**	**$12,619 vs $1,080 | +4.0 pts**	**High**
4	**An 8.7-point gap exists between the highest and lowest-scoring physicians**	**78.5 (Dr Johnson) to 69.8 (Dr Smith)**	**High**
5	Adults aged 30–44 score 2.1 pts below average — the lowest major age group	Score 73.2 | n=71 | avg cost $3,919	Medium
6	Dermatology leads on recovery, cost, and discharge speed combined	76.6 | $4,728 | 4.7 days	Opportunity
7	Oncology achieves above-average recovery at the second-lowest departmental cost	75.1 | $4,646	Opportunity

5B. EXECUTIVE RISK & OPPORTUNITY HEATMAP

Operational area	Recovery	Efficiency	Financial	Priority
Cardiology	**High risk (72.9)**	**High risk (5.6d)**	**High ($5,255/pt)**	**IMMEDIATE**
Observation pathway	**Below avg (72.9)**	**Moderate risk**	**Low ($1,080)**	**HIGH**
Dr Smith & Dr Rodriguez	**69.8 / 72.8**	**Moderate risk**	**Elevated cost**	**HIGH**
Adults (30–44)	Below avg (73.2)	Moderate risk	Long-term impact	MEDIUM
General Medicine	Strong (76.6)	Capacity (5.9d)	Moderate ($5,157)	Optimise
Dermatology	Best (76.6)	Best (4.7d)	Low ($4,728)	Replicate
Oncology	Above avg (75.1)	Strong (4.8d)	Low ($4,646)	Replicate

5C. EXECUTIVE RECOMMENDATIONS

Each recommendation is tied to a specific data finding. All six include a measurable target and timeframe so progress can be reviewed at the next reporting cycle.

1. Review Cardiology's operational model within 60 days [IMMEDIATE]

Cardiology admissions cost an average of $5,255 and produce a recovery score of 72.9 — the lowest in the hospital. A formal workflow review should examine how patients are stabilised on admission, how discharge decisions are made, and whether the current staffing model matches patient complexity. Use Dermatology as the internal comparison: $4,728 average cost, 76.6 recovery, and the hospital's shortest average stay of 4.7 days.

Target: Cardiology recovery score reaches 74.5 within two reporting cycles

2. Set a 48-hour escalation policy for Observation patients [HIGH]

Patients on Observation average 72.9 across 102 admissions — the lowest of any treatment track. The hospital should require that stable Observation patients are reviewed for escalation to an active treatment within 24 to 48 hours. Therapy handles a similar volume (99 cases) and produces a 76.1 recovery score at $3,954.

Target: Observation pathway recovery score reaches 74.5 within two reporting cycles

3. Put Dr Smith and Dr Rodriguez into a structured peer review [HIGH]

Dr Smith's patients average 69.8 across 27 admissions — 5.6 points below average. His average cost of $5,756 is the second-highest among all physicians. Dr Rodriguez sees the most patients (55 cases) at 72.8. Pair both with Dr Johnson (78.5, n=47) and Dr Jones (76.9, n=43).

Target: All physicians reach a recovery score above 73.0 within three reporting cycles

4. Investigate and support the adult patient cohort specifically [HIGH]

Adults aged 30 to 44 average 73.2 across 71 admissions — 2.1 points below average — at the lowest average cost of any age band ($3,919). The hospital should review whether this cohort is placed on lower-intensity pathways that do not match clinical needs, and examine post-discharge follow-up availability.

Target: Adult cohort recovery score reaches 75.0 within four reporting cycles

5. Track every high-cost admission against its actual recovery score [MEDIUM]

81 patients — 20.3% of all admissions — are in the high-cost category averaging $13,877. A monthly review should flag any high-cost admission scoring below 75.3 for departmental review, with

Cardiology reviewed first.

Target: Reduce proportion of high-cost admissions scoring below the hospital average within six cycles

6. Document and share Dermatology and Oncology's operating practices [MEDIUM]

Dermatology (76.6 recovery, $4,728 cost, 4.7-day stay) and Oncology (75.1 recovery, $4,646 cost, 4.8-day stay) both achieve above-average results at below-average cost. Leadership should ask these departments to document their treatment sequencing, discharge planning, and resource allocation practices, then share those findings with Cardiology and General Medicine.

Target: Benchmark review completed within 90 days; at least one practice piloted in Cardiology within two cycles

5D. FUTURE ANALYTICS & CLOSING NOTE

Where this analysis can go next

- Predict recovery scores for new patients based on department, age group, and treatment type
- Forecast readmission risk using discharge scores and length-of-stay data
- Build a live dashboard connected directly to the hospital's patient management system
- Model physician workload against patient outcomes to find the right caseload per doctor
- Move the dashboard to Power BI or Tableau for enterprise-scale reporting
- Use machine learning to flag clinical anomalies before they affect outcomes

Closing note

This analysis took 400 records that were sitting in raw tables and turned them into six decisions leadership can act on this quarter. The data shows a hospital with strong departments — particularly Dermatology and General Medicine, alongside a clear majority of above-average physicians — and several areas that merit closer review, especially Cardiology, the Observation pathway, and a small number of physicians whose patient outcomes fall below the institutional norm.

The most important finding is not simply that Cardiology is expensive — it is that Cardiology is expensive while also performing below the hospital average in this dataset. That combination makes Cardiology the clearest priority for review; the workflow audit recommended below is intended to determine whether the pattern reflects a process issue, case-mix differences, or both. Acting on Cardiology, escalating Observation patients faster, and narrowing the physician performance gap could, if these reviews confirm the patterns, improve the hospital's overall recovery average in a future reporting cycle.

Final summary for executives

Two departments lead the hospital and should be studied further: Dermatology and Oncology. Two areas warrant immediate review: Cardiology and the Observation treatment pathway. Two physicians would benefit from structured support now: Dr Smith and Dr Rodriguez. One demographic group may need a dedicated follow-up programme: adults aged 30 to 44. All six recommendations come with targets and timelines, and the next review cycle should measure progress against each one.

A. APPENDIX — FULL DASHBOARD SNAPSHOT

The dashboard below is the complete interactive Excel dashboard as submitted. It features five KPI tiles, six pivot charts, and four slicer panels — Department, Gender, Doctor Name, and Treatment Type.

Full Dashboard Snapshot

Figure Dashboard — Full Excel Dashboard Screenshot

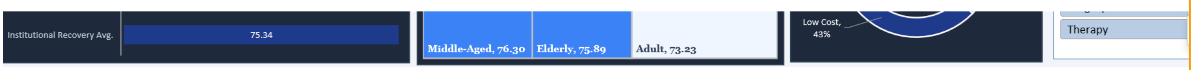

Figure Dashboard: *Green Valley Medical Center — Executive Healthcare Intelligence Dashboard. Five KPI tiles, six pivot charts, four interactive slicer panels.*